

HI BUDDY BE SUPER HONEST DON'T COMPROMISE ANYTHING TELL ME 5 YEAR EXPERIENCE LEVEL OF DATA ENGINEER 60 LPA PACKAGE WORTH PERSON SKILLSET PLEASE GATHER ALL TRENDING AND PERFECT SKILLSET DON'T COMPROMISE ANYTHING FROM ALL DB LIKE POSTGRESS MONGODB AND ALL AND ALL TO MS FABRIC INCLUD EVERYTHIN AND GIEV ME PERFECT SYLLABUS WITH EMOJI

ChatGPT gpt-5-5

Absolutely. If your target is **₹60 LPA+ as a Data Engineer with ~5 years of experience**, then don't think in terms of "learning tools." Think in terms of becoming someone who can **design, build, optimize, secure, and operate data platforms at scale**.

The engineers getting ₹50–80 LPA in companies like Netflix, Uber, Atlassian, Microsoft, Google, Amazon, Rubrik, Adobe, Walmart Global Tech, Snowflake, Databricks, Salesforce, and high-growth startups typically have **depth in fundamentals + breadth across the modern data stack**.

Data Engineer Roadmap (5 Years Experience | ₹60+ LPA Level)

Think of this as **Data Engineering University**.

PHASE 1 — Programming Mastery

Python (Expert)

Core Python

- Variables
- Loops
- Functions
- OOP
- Exception Handling
- File Handling
- Modules
- Decorators
- Generators
- Iterators
- Context Managers

Advanced Python

- Multithreading

- Multiprocessing
- AsyncIO
- Type Hinting
- Dataclasses
- Logging
- Profiling
- Memory Management
- Packaging
- Virtual Environments
- Pytest
- Ruff
- Black
- Poetry

Libraries

- ✓ Pandas
- ✓ NumPy
- ✓ Polars
- ✓ PyArrow
- ✓ Requests
- ✓ BeautifulSoup
- ✓ Selenium
- ✓ FastAPI
- ✓ SQLAlchemy
- ✓ Pydantic
- ✓ OpenPyXL
- ✓ Faker
- ✓ boto3
- ✓ Azure SDK
- ✓ Google Cloud SDK

SQL (Expert Level)

Not intermediate.

Expert.

DDL

- CREATE
 - ALTER
 - DROP
-

DML

- INSERT
 - UPDATE
 - DELETE
 - MERGE
-

Queries

- GROUP BY
 - HAVING
 - EXISTS
 - IN
 - ANY
 - ALL
-

Advanced SQL

Window Functions

- ROW_NUMBER
 - DENSE_RANK
 - LAG
 - LEAD
 - FIRST_VALUE
 - LAST_VALUE
-

CTEs

Recursive Queries

Pivot

Unpivot

Dynamic SQL

Materialized Views

Indexes

Partitioning

Execution Plans

Query Optimization

Statistics

Transactions

Isolation Levels

Locks

Deadlocks

Normalization

Denormalization

Star Schema

Snowflake Schema

Slowly Changing Dimensions (SCD Types 0–6)

Relational Databases

Become comfortable with all major relational systems:

- PostgreSQL ★★★★★
- MySQL
- MariaDB
- Microsoft SQL Server
- Oracle Database
- SQLite (lightweight use cases)

Know:

- Installation
 - Backup & Restore
 - Replication
 - Partitioning
 - Indexing
 - Stored Procedures
 - Functions
 - Triggers
 - Views
 - Performance Tuning
 - Security
 - High Availability
-

MongoDB

- Aggregation Pipeline
 - Sharding
 - Replica Sets
 - Indexes
 - Transactions
 - Schema Design
 - Performance Tuning
-

Cassandra

- Partition Keys
 - Clustering Keys
 - Data Modeling
 - Consistency Levels
 - Compaction
 - Repair
-

Redis

- Cache
 - Pub/Sub
 - Streams
 - TTL
 - Persistence
-

DynamoDB

- Partition Key
 - Sort Key
 - GSIs
 - LSIs
 - Capacity Modes
-

Elasticsearch / OpenSearch

- Indexes
- Mapping
- Search

- Aggregations
- Performance

Neo4j (Graph)

- Cypher
- Nodes
- Relationships

Time-Series Databases

- InfluxDB
- TimescaleDB

File Formats

Understand when and why to use:

- ✓ CSV
- ✓ JSON
- ✓ XML
- ✓ Avro
- ✓ ORC
- ✓ Parquet ★★★★★
- ✓ Delta Lake ★★★★★
- ✓ Iceberg ★★★★★
- ✓ Hudi ★★★★★

Big Data

Hadoop

- HDFS
- YARN
- MapReduce

Spark (Expert)

RDD

DataFrame

Dataset

Spark SQL

Spark Streaming

Broadcast Variables

Accumulators

Catalyst Optimizer

AQE

Caching

Checkpointing

Shuffle

Partitioning

Performance Tuning

PySpark

Spark Scala basics

Streaming

Kafka ★★★★★

Topics

Partitions

Replication

Offsets

Consumer Groups

Transactions

Schema Registry

Avro

Kafka Connect

MirrorMaker

Streams API

Apache Flink

Spark Streaming

Azure Event Hubs

AWS Kinesis

ETL / ELT

Design patterns:

- Batch
 - Streaming
 - CDC
 - Incremental Loading
 - Full Load
 - Merge
 - Upsert
 - Watermarking
 - Idempotency
 - Error Handling
 - Retry
 - DLQ
 - Data Validation
 - Reconciliation
-

Data Warehousing

Kimball

Inmon

Data Vault 2.0

Fact Tables

Dimension Tables

Surrogate Keys

Conformed Dimensions

Star Schema

Snowflake Schema

Bridge Tables

Junk Dimensions

Mini Dimensions

Late Arriving Dimensions

Cloud Platforms

Azure ★★★★★

Storage Account

ADLS Gen2

Blob Storage

Key Vault

Azure SQL

Azure Functions

Synapse Analytics

Azure Databricks

Azure Data Factory

Microsoft Fabric ★★★★★

Azure Monitor

Azure DevOps

Event Hubs

Service Bus

Logic Apps

Managed Identity

RBAC

Private Endpoints

AWS

S3

Glue

Athena

EMR

Redshift

RDS

Lambda

IAM

Lake Formation

MSK

Kinesis

Step Functions

CloudWatch

Google Cloud

BigQuery

Cloud Storage

Dataflow

Dataproc

Composer

Pub/Sub

Cloud Run

Cloud Functions



Microsoft Fabric (Expert)

- OneLake
 - Lakehouse
 - Warehouse
 - Data Factory
 - Data Pipelines
 - Dataflows Gen2
 - Notebooks
 - Spark
 - Mirroring
 - Eventstream
 - KQL Database
 - Real-Time Intelligence
 - Semantic Models
 - Power BI Integration
 - Shortcuts
 - Deployment Pipelines
 - Git Integration
 - Security
 - Capacity
 - Governance
 - Monitoring
 - Performance Optimization
-

Orchestration

Apache Airflow ★★★★★

- DAGs
- Sensors
- Executors
- Scheduling
- XCom
- Variables
- Connections
- Dynamic DAGs
- TaskFlow API
- Backfill
- SLA
- Retry Strategy

Also know:

- Prefect
- Dagster
- Azure Data Factory Pipelines
- Microsoft Fabric Pipelines

Data Lake

Learn:

Bronze

Silver

Gold

Medallion Architecture

Lakehouse

Delta Lake

Iceberg

ACID

Time Travel

Vacuum

Optimization

Z-Ordering

Analytics

Power BI ★★★★★

DAX

Power Query

Semantic Model

RLS

Incremental Refresh

Deployment Pipelines

Fabric Integration

Tableau

Looker

Apache Superset

APIs

REST

GraphQL

OAuth

JWT

Pagination

Rate Limiting

Webhooks

OpenAPI / Swagger

DevOps

Git ★★★★★

GitHub

GitLab

Azure DevOps

CI/CD

Docker

Docker Compose

Kubernetes Basics

Terraform

Bicep (Azure)

Helm (nice to have)

Data Quality

Great Expectations

Soda

Deequ

dbt Tests

Custom Validation Frameworks

dbt (Expert)

Models

Seeds

Snapshots

Macros

Tests

Sources

Documentation

Exposures

Incremental Models

Packages

Monitoring

Prometheus

Grafana

Azure Monitor

CloudWatch

Datadog

OpenTelemetry

ELK Stack

Security

RBAC

ABAC

Encryption at Rest

Encryption in Transit

Secrets Management

Key Vault

IAM

Least Privilege

GDPR

PII

Data Masking

Row-Level Security

Column-Level Security

Auditing

Software Engineering

Design Patterns

SOLID

Clean Code

Testing

Unit Testing

Integration Testing

Mocking

Logging

Exception Handling

Code Reviews

Documentation

System Design

Data Pipeline Design

Real-Time Systems

CDC Architecture

Event-Driven Architecture

Lambda Architecture

Kappa Architecture

Lakehouse Design

Warehouse Design

Scalability

Cost Optimization

Disaster Recovery

High Availability



AI for Data Engineers

Prompt Engineering

LLMs

Vector Databases

RAG

Embeddings

AI-assisted ETL

Copilot tools

MCP concepts

AI agents that interact with data systems



Mathematics & Statistics

Probability

Statistics

Sampling

Percentiles

Distributions

Basic Linear Algebra

Data Quality Metrics



Business Knowledge

Finance

Retail

Healthcare

Insurance

Manufacturing

Supply Chain

Telecom

Marketing Analytics

Knowing how data supports business decisions is often what separates senior engineers from mid-level engineers.

Soft Skills

- Architecture discussions
 - Technical documentation
 - Estimation
 - Stakeholder communication
 - Mentoring
 - Incident management
 - Root cause analysis
 - Agile / Scrum
 - Technical leadership
-

Portfolio (Non-Negotiable)

Build projects that resemble production systems:

-  End-to-end batch ETL pipeline
 -  Real-time Kafka → Spark → Delta Lake pipeline
 -  Azure Data Factory → ADLS → Databricks → Power BI
 -  Microsoft Fabric Lakehouse with Medallion architecture
 -  dbt transformations with automated tests
 -  CDC pipeline from PostgreSQL to a data warehouse
 -  Data lake using Delta Lake or Iceberg
 -  Monitoring dashboard with Grafana/Prometheus
 -  Secure cloud deployment with CI/CD
 -  AI-powered data quality or metadata assistant
-

What actually differentiates a ₹60 LPA engineer?

By 5 years, you should be able to:

-  Design an end-to-end enterprise data platform.
-  Optimize Spark and SQL jobs handling terabytes of data.
-  Choose the right database based on workload and access patterns.

-  Build reliable batch and streaming pipelines with proper observability.
-  Implement CI/CD, testing, security, and infrastructure as code.
-  Lead technical discussions, mentor engineers, and review architecture.
-  Balance performance, scalability, reliability, governance, and cloud costs.

★ Priority (80/20)

If time is limited, focus deepest on:

1.  Python
2.  SQL (expert level)
3.  PostgreSQL
4.  MongoDB
5.  Apache Spark (PySpark)
6.  Kafka
7.  Data Warehousing & Dimensional Modeling
8.  Azure (ADLS, Data Factory, Databricks)
9.  Microsoft Fabric
10.  Delta Lake & Lakehouse Architecture
11.  Apache Airflow
12.  dbt
13.  Docker + Git + CI/CD
14.  Power BI
15.  Security, Governance & System Design

Mastering this list deeply will make you competitive for senior data engineering roles. The remaining technologies broaden your versatility and help you adapt to different companies' stacks, but the biggest salary gains come from strong fundamentals, distributed systems knowledge, cloud expertise, and the ability to design and operate production-grade data platforms.

2026-08-03 23:56:57

You Asked 

SO IF ONE WANT TO LEARN ALL THEN TELL ME HOW MANY HR ONE NEEDED FOR THAT

 ChatGPT gpt-5-5

If someone starts from **zero** and wants to genuinely master the entire roadmap (not just watch videos), the time investment is much larger than most people expect.

Here's a realistic estimate:

Level	Hours	Outcome
🟢 Basics	800–1,200 hrs	Junior Data Engineer
🟡 Intermediate	1,800–2,500 hrs	2–3 years experience equivalent
🟣 Advanced	3,500–4,500 hrs	Strong Senior Data Engineer
🟦 Expert (₹50–60+ LPA level)	5,000–7,000 hrs	Can design and lead enterprise-scale data platforms

Important: A ₹60 LPA engineer is not paid just for knowing tools. They're paid for solving difficult production problems, making architectural decisions, and delivering reliable systems.

 Approximate breakdown

Skill Area	Hours
🐍 Python	350
🗄️ SQL (Expert)	350
🐘 PostgreSQL & SQL Server	250
🌿 MongoDB + NoSQL	200
⚡ Spark / PySpark	500
🌀 Kafka & Streaming	300
🏗️ Data Warehousing	250
☁️ Azure Cloud	500
📦 Microsoft Fabric	350
🔄 Airflow	180
🌳 dbt	120
📦 Delta Lake / Iceberg	180
🐳 Docker + Kubernetes	250
🚀 CI/CD + DevOps	200
📊 Power BI	200
🔒 Security & Governance	180
🏛️ System Design	400
🤖 AI for Data Engineers	250
👛 Real Projects (10–15)	1,500+
🧠 Interview Prep & DSA (basic)	300

Total: ≈ 5,500–6,500 focused hours

How long does that take?

Daily Study	Time to Reach ~6,000 Hours
2 hrs/day	~8.2 years
4 hrs/day	~4.1 years
6 hrs/day	~2.7 years
8 hrs/day	~2.1 years
10 hrs/day	~1.6 years

These assume **consistent, high-quality practice every day**, not just passive video watching.

Reality Check

Many people spend:

-  2,000 hours watching tutorials
-  Only 200 hours actually building projects

That rarely leads to senior-level skills.

A stronger ratio is:

- **30% learning** 
- **70% hands-on building** 

If you want to become truly elite

Aim for approximately:

-  1,500 hours learning concepts
-  3,000 hours building real projects
-  1,000 hours debugging production-like issues
-  500 hours on system design and interview preparation

That combination is much closer to what creates a genuinely strong senior data engineer than simply collecting certificates.

If you're willing to invest **5,000–7,000 focused hours**, you'll be in the range of the strongest data engineers—assuming those hours are spent deliberately on building, debugging, optimizing, and understanding systems, not just consuming content.

2026-08-04 14:09:24 You Asked 

Ok one can spend 6 hr for 200 days so now tell me the plan for that

Yes. Let's do the math first.

Your Commitment

-  200 Days
-  6 Hours/Day

Total = 1,200 Hours

You **cannot master everything** in 1,200 hours. Anyone who says you can is being unrealistic.

However, **1,200 focused hours is enough to become a strong Data Engineer** capable of landing many good roles if you spend those hours wisely.

200-Day Elite Data Engineering Plan (1,200 Hours)

text

Total Hours = 1,200

Python	180 hrs
SQL	180 hrs
PostgreSQL	60 hrs
MongoDB	50 hrs
Data Modeling	50 hrs
Spark/PySpark	160 hrs
Azure Cloud	120 hrs
Microsoft Fabric	100 hrs
Airflow	50 hrs
Kafka	60 hrs
Power BI	40 hrs
Git + Docker	40 hrs
CI/CD	30 hrs
System Design	40 hrs
Projects	180 hrs
Interview Prep	60 hrs

TOTAL = 1,200 Hours

Phase 1 (Days 1–40) – Programming & SQL Foundation (240 Hours)

Python (120 Hours)

Core Python

- Variables
- Data Types
- Loops
- Functions
- OOP

- File Handling
- Exceptions
- Modules
- Virtual Environments

Advanced

- Decorators
- Generators
- Iterators
- Context Managers
- Logging
- Multithreading
- Async Basics
- Type Hinting

Libraries

- Pandas
- NumPy
- Requests
- SQLAlchemy
- FastAPI (Basics)

SQL (120 Hours)

Master

- SELECT
- JOIN
- GROUP BY
- HAVING
- CASE
- CTE
- Window Functions
- Recursive Queries
- Indexing
- Transactions
- Query Optimization
- Execution Plans

^{July} 17 Phase 2 (Days 41–70) – Databases (180 Hours)

PostgreSQL

- Installation
 - Indexes
 - Views
 - Triggers
 - Functions
 - Partitioning
 - Backup
 - Restore
-

MongoDB

- CRUD
 - Aggregation Pipeline
 - Indexes
 - Replica Sets
 - Sharding
-

Data Modeling

Star Schema

Snowflake Schema

Fact Tables

Dimension Tables

SCD

Normalization

Denormalization

Phase 3 (Days 71–105) – Big Data (210 Hours)

Spark

- Architecture
 - DataFrames
 - Spark SQL
 - UDFs
 - Partitioning
 - Caching
 - Performance
 - Joins
 - Optimization
-

PySpark

Build

- ETL
 - Incremental Loading
 - Delta Tables
 - Batch Jobs
-

Phase 4 (Days 106–140) — Azure + Fabric (210 Hours)

Azure

- Storage
 - ADLS
 - Azure SQL
 - Data Factory
 - Databricks
 - Synapse
 - Key Vault
-

Microsoft Fabric

- OneLake
 - Lakehouse
 - Warehouse
 - Pipelines
 - Notebooks
 - Spark
 - Semantic Models
 - Deployment
 - Monitoring
-

Phase 5 (Days 141–165) — Streaming + Orchestration (150 Hours)

Kafka

- Topics
- Producers
- Consumers
- Consumer Groups
- Offsets
- Schema Registry

Airflow

- DAGs
 - Scheduling
 - Retry
 - Sensors
 - Dynamic DAGs
-

Git

- Branches
 - Merge
 - Rebase
 - Pull Requests
-

Docker

- Dockerfile
 - Compose
 - Containers
-

Phase 6 (Days 166–200) — Real Projects (210 Hours)

Project 1

Enterprise ETL

Python

Postgres

Airflow

Azure Storage

Project 2

Streaming

Kafka

Spark

Delta Lake

Power BI

Project 3

Azure Data Factory

ADLS

Databricks

Fabric

Project 4

Fabric End-to-End

OneLake

Warehouse

Power BI

Deployment

Project 5

Production Pipeline

Docker

CI/CD

Monitoring

Logging

Alerts

Daily Schedule (6 Hours)

Hour 1

 Learn theory

Hour 2

 Code along

Hour 3

 Practice from scratch

Hour 4

 Build a mini-project

Hour 5

 Debug & optimize

Hour 6

Weekly Plan

Monday–Friday (30 hrs):

- Learn new concepts
- Solve practice exercises
- Build mini-projects

Saturday (6 hrs):

- Complete one end-to-end feature
- Refactor code
- Write documentation

Sunday (6 hrs):

- Revision
 - Mock interview
 - SQL challenge
 - Read engineering blogs
 - Update GitHub portfolio
-

Milestones

Day 40

- Strong Python
- Strong SQL

Day 70

- PostgreSQL + MongoDB
- Data Modeling

Day 105

- PySpark
- ETL Pipelines

Day 140

- Azure + Microsoft Fabric

Day 165

- Kafka
- Airflow
- Docker
- Git

✓ Day 200

- 5 production-style projects on GitHub
- Portfolio ready
- Resume ready
- Interview-ready for mid-level to senior-leaning data engineering roles

🚀 If your long-term goal is ₹60 LPA

Think of this **200-day plan as Phase 1**. It gives you a very strong foundation (about **1,200 hours**).

Then continue with **another 1,200–2,000 hours** over the following year focusing on:

- 🏢 Advanced system design
- ⚡ Spark performance tuning at scale
- ☁ Advanced cloud architecture
- 🛡 Security and governance
- 🌐 Multi-cloud (Azure + AWS/GCP)
- 🔧 Production debugging and observability
- 📈 Leading larger, enterprise-grade projects

That second phase is what typically separates a competent engineer from someone who consistently earns **₹60 LPA+**.

2026-08-04 14:10:15 You Asked 

So if I learn all this what I can get then

ChatGPT gpt-5-5

If you **actually master** the 1,200-hour roadmap—not just complete courses, but build projects, understand the concepts deeply, and can explain your decisions—you'll be in a strong position. Here's a realistic picture.

🎯 After 1,200 Hours, You Can Become

✓ Job-ready for:

- 💼 Data Engineer
- ☁ Azure Data Engineer
- 🏢 Microsoft Fabric Data Engineer
- 📊 Analytics Engineer (with dbt/Power BI if you add those)
- ⚡ ETL Developer
- 🚀 Big Data Engineer (entry to mid-level)

💰 Expected Salary (India)

These are **approximate ranges** and depend heavily on your existing experience, communication, projects, and interview performance.

Profile	Approximate Salary
Fresher with strong projects	₹6–12 LPA
1–2 years experience	₹10–18 LPA
3 years experience	₹15–25 LPA
5 years with strong skills	₹25–45+ LPA
Top product companies with exceptional skills	₹40–60+ LPA

A ₹60 LPA package usually requires **both** strong technical skills **and** relevant professional experience. Learning alone is rarely enough.

🏆 What You Will Be Able to Build

You'll be capable of creating systems like:

- 🔗 End-to-end ETL pipelines
- ☁ Azure Data Factory → ADLS → Databricks → Fabric workflows
- ⚡ Kafka + Spark real-time pipelines
- 🗄 Data warehouses with star schemas
- 📊 Power BI dashboards connected to Fabric
- 🔄 Airflow-orchestrated pipelines
- 🐳 Dockerized data applications
- 📦 Production-ready GitHub projects

These are the kinds of projects employers look for because they resemble real work.

🎤 Interview Readiness

You should be able to answer questions such as:

- Why choose PostgreSQL instead of MongoDB?
- When should you partition a Spark DataFrame?
- How does Kafka guarantee message delivery?
- How would you design a data lakehouse?
- How do you optimize a slow SQL query?
- How do you implement incremental loading?
- How would you build a production ETL pipeline?
- How would you secure data in Azure?

If you can confidently answer and demonstrate these topics, you'll stand out.

What Comes Next?

The 1,200-hour plan gets you to a **strong professional level**, but the engineers earning **₹60–80+ LPA** usually add:

-  Advanced system design
-  Spark optimization on large datasets
-  Multi-cloud experience (Azure + AWS/GCP)
-  Observability and monitoring
-  Enterprise security and governance
-  Technical leadership and mentoring
-  Experience solving real production incidents

My assessment

If someone completes the roadmap with **real understanding**, builds **5–10 high-quality production-style projects**, has **good communication**, and performs well in interviews, they can become one of the stronger data engineering candidates in the market.

The biggest factor after that isn't another certificate—it's **real-world experience applying those skills at scale**. That's what typically moves compensation from good to exceptional.

2026-08-04 14:10:58

You Asked 

18 lpa I hand

ChatGPT gpt-5-5

If you mean "**Is ₹18 LPA in-hand?**", the answer is:

No. ₹18 LPA is your **CTC (Cost to Company)**, not the amount you receive in your bank account.

A rough estimate for a typical ₹18 LPA CTC in India:

-  **CTC:** ₹18,00,000/year
-  **Monthly in-hand:** approximately **₹1.10–1.35 lakh/month**

The exact amount depends on:

- Basic salary structure
- PF deductions
- Income tax regime (old/new)
- Variable pay or bonuses
- ESOPs
- Gratuity
- Employer benefits (insurance, etc.)

If you meant something else:

- **Can you get ₹18 LPA after completing the roadmap?** → **Yes, it's possible**, especially if you already have experience or build a strong portfolio and perform well in interviews.
- **As a fresher:** ₹18 LPA is achievable but not common. It usually requires exceptional skills and landing a top product company or a high-paying startup.
- **With 2–5 years of relevant experience and the skills you described:** ₹18 LPA is a realistic target.

If you're asking about **salary progression**, a strong path could look like:

- 🚀 Entry: ₹8–15 LPA
- 💼 Mid-level: ₹15–25 LPA
- ★ Senior: ₹25–45 LPA
- 🏆 Top product companies: ₹45–60+ LPA (and higher in some cases)

If your goal is **₹60 LPA**, I can create a **day-by-day 200-day roadmap** (every single day planned for 6 hours) to maximize your chances.

2026-08-06 11:24:57 You Asked 

go for it

 ChatGPT gpt-5-5

I love this goal. If you're willing to put in **6 focused hours every day for 200 days**, I'll treat this like a **bootcamp for becoming a top-tier Data Engineer**, not just someone who collects certificates.

🎯 End Goal (Day 200)

By the end of 200 days, you should have:

- 🐍 Expert-level Python for data engineering
- 🗄️ Advanced SQL
- 🐘 PostgreSQL
- 🟢 MongoDB
- ⚡ PySpark
- 🌊 Kafka
- ☁️ Azure
- 🏢 Microsoft Fabric
- 🔗 Airflow
- 🌳 dbt
- 🐳 Docker + Git + CI/CD
- 📊 Power BI

- 🏗️ 8–10 enterprise-quality projects
- 📁 Resume ready
- 🌐 GitHub portfolio
- 🎤 Interview ready

July 17

Master Plan

Phase	Days	Focus
🌱 Phase 1	1–30	Python
📚 Phase 2	31–60	SQL + PostgreSQL
💧 Phase 3	61–75	MongoDB + Data Modeling
⚡ Phase 4	76–105	Spark + PySpark
☁ Phase 5	106–135	Azure
🏢 Phase 6	136–155	Microsoft Fabric
🌊 Phase 7	156–170	Kafka + Airflow
🚀 Phase 8	171–200	Projects + Interviews

🕒

Daily Schedule (6 Hours)

📖

Hour 1

Theory

👨💻

Hour 2

Coding

🔥

Hour 3

Hands-on Practice

🔧

Hour 4

Mini Project

🐛

Hour 5

Debugging & Exercises

📝

Hour 6

Revision + GitHub + Notes

🌱

PHASE 1 (Days 1–30)

Week 1

Day 1

- Python Installation
- VS Code
- Git
- Variables
- Data Types

Mini Project

Calculator

Day 2

- Loops
- Functions

Mini Project

Student Grade System

Day 3

- Lists
- Tuples
- Dictionaries
- Sets

Mini Project

Inventory Manager

Day 4

- File Handling
- JSON
- CSV

Mini Project

Employee Data Reader

Day 5

- Exception Handling
- Logging

Mini Project

Bank Application

Day 6

- OOP

Mini Project

Library Management System

Day 7

Revision

Python Quiz

GitHub Upload

Week 2

Advanced Python

- Decorators
- Generators
- Lambda
- Iterators
- Context Managers

Project

Log Analyzer

Week 3

Pandas

NumPy

Requests

Project

COVID Dataset Analysis

Week 4

FastAPI

SQLAlchemy

API Development

Project

Employee REST API

PHASE 2 (Days 31–60)

Advanced SQL

Every day

- 40 SQL Questions

Topics

- Joins
- Window Functions
- CTE
- Recursive SQL
- Transactions
- Indexing
- Query Optimization

Projects

Hospital Database

Bank Database

Retail Database

PostgreSQL

Install

Backup

Restore

Functions

Triggers

Partitioning

Performance

PHASE 3 (Days 61–75)

MongoDB

Aggregation

Replica Sets

Indexes

Project

E-Commerce Database

Data Modeling

Star Schema

Snowflake

SCD

Fact Tables

Dimension Tables

Project

Retail Warehouse

PHASE 4 (Days 76–105)

Spark

PySpark

RDD

DataFrame

Spark SQL

Partitioning

Caching

Optimization

Projects

Sales ETL

Movie Recommendation Data Pipeline

Delta Lake Pipeline

PHASE 5 (Days 106–135)

Azure

- Storage Account
- ADLS
- Azure SQL
- Key Vault
- Data Factory
- Databricks
- Synapse

Projects

Azure ETL

 PHASE 6 (Days 136–155)

Microsoft Fabric

- OneLake
- Lakehouse
- Warehouse
- Data Factory
- Spark
- Pipelines
- Semantic Models

Projects

Enterprise Fabric Pipeline

Lakehouse Analytics

 PHASE 7 (Days 156–170)

Kafka

Topics

Consumers

Partitions

Schema Registry

Project

Real-Time Orders Pipeline

Airflow

DAG

Scheduling

Retries

Project

Daily ETL Automation

Docker

Dockerfile

Compose

Containerization

PHASE 8 (Days 171–200)

Enterprise Projects

Project 1

Retail Data Warehouse

Tech

Python

PostgreSQL

Airflow

Azure

Power BI

Project 2

Netflix Analytics

PySpark

Delta Lake

Azure

Project 3

Real-Time Banking

Kafka

Spark

MongoDB

Project 4

Fabric End-to-End

OneLake

Warehouse

Semantic Model

Power BI

Project 5

IoT Analytics

Kafka

Spark Streaming

Fabric

Project 6

Healthcare Data Lake

ADLS

Databricks

Delta

Project 7

Fraud Detection Data Pipeline

Kafka

Spark

PostgreSQL

Project 8

HR Analytics

Azure Data Factory

Fabric

Power BI

Interview Preparation (Daily During Last 30 Days)

Every day:

-  20 SQL problems
 -  2 Python coding problems
 -  1 Spark interview topic
 -  1 System Design discussion
 -  Explain one project aloud in 10–15 minutes
 -  Mock interview questions
-

Books

-  *Python Crash Course*
-  *Fluent Python*
-  *Designing Data-Intensive Applications* (advanced)
-  *Spark: The Definitive Guide*
-  *Fundamentals of Data Engineering*

GitHub Portfolio (Minimum)

By Day 200, your GitHub should contain:

- 🐍 30 Python repositories
 - 🗄️ 50 SQL scripts
 - 🐘 PostgreSQL project
 - 🍃 MongoDB project
 - ⚡ 5 Spark projects
 - ☁️ 5 Azure projects
 - 🏢 3 Microsoft Fabric projects
 - 🗄️ Kafka project
 - 🔗 Airflow project
 - 📊 Power BI dashboards
 - 🐳 Dockerized applications
 - 📄 Clear README files with architecture diagrams and setup instructions
-

Beyond Day 200

This plan gets you to a strong professional level, but **₹60 LPA** usually comes after **applying these skills in real production environments**. After these 200 days, spend the next 1–3 years deepening your expertise by:

- Optimizing Spark jobs handling large datasets
- Designing scalable data architectures
- Solving production incidents
- Learning cloud cost optimization
- Mentoring teammates and leading technical initiatives

One recommendation

Instead of trying to memorize every technology, focus on understanding **why** each one exists, **when** to use it, and **what trade-offs** it has. That mindset is what distinguishes senior engineers from people who simply know many tools.

I can also build this into a detailed 200-day curriculum with all 1,200 hours planned—every day including topics, exercises, project tasks, resources, and interview questions—so it feels like a complete engineering course rather than just a roadmap.